

DEREK CHIBUZOR

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EDUCATION

University of Southern California	Los Angeles, CA
Master of Science in Mechanical Engineering	Aug. 2025-Dec. 2026
Bachelor of Science in Aerospace Engineering	Aug. 2021-May 2025

EXPERIENCE

Neros Technologies	Torrance, CA
Controls Engineer, Autonomy	Jan. 2026-Present

- * Develop guidance, navigation, and control algorithms supporting core autonomy capabilities for compute-constrained UAVs.
- * Contribute to internal 6-DoF flight simulator with custom dynamics, control, state estimation, and perception modules.
- * Design and conduct experiments to tune controllers, characterize sensor performance, and evaluate closed-loop system behavior.

Dynamic Robotics and Control Laboratory	Los Angeles, CA
Research Assistant, Sim2Real	Jan. 2025-Present

- * Engineered system identification framework for Sim2Real transfer via gradient- and sampling-based hybrid optimization.
- * Trained evolutionary algorithms and machine learning models to improve actuator dynamics modelling by 78% in 24-DoF robot.
- * Constructed diffusion model framework for generating full-body loco-manipulation trajectories from Unitree G1 motion capture data.

Lawrence Livermore National Laboratory	Livermore, CA
Computational Engineer Intern	June 2025-Aug. 2025

- * Wrote IK task-space PD controller for 6-DoF UR3e, saving 5+ hours of manual path planning while achieving 0.1 mm tracking error.
- * Trained reinforcement learning policies for pick and place tasks, reducing episode length by 20% with transformer feature extractor.
- * Utilized implicit and explicit structural dynamics codes to perform modal analysis of jointed beam subject to broadband excitation.

Northrop Grumman	Roy, UT
Mechanical Engineer Intern	June 2023-Aug. 2024

- * Constructed 2-DoF reduced-order Simulink model of subterranean shock-isolated system, saving 240 hours of FEA computation.
- * Designed CAD assemblies, performed FEA, and wrote RFIs for elastomeric shock isolators and maintenance access doors.

Amazon & Information Sciences Institute	Los Angeles, CA
GNC Engineer Intern	June 2022-Aug. 2022

- * Developed vision-based navigation software, enabling pose estimation for 3-DoF rendezvous and proximity operations (RPO).
- * Achieved sub-160 ms latency localization and tracking of 12 infrared LED targets up to 2.50 m from Raspberry Pi NoIR camera.

SKILLS

Languages: Python, C++, MATLAB, Java, JavaScript

Software: Simulink, Siemens NX, PyTorch, MuJoCo, CasADi, JAX, OpenCV, LabVIEW, Betaflight

Hardware: Raspberry Pi CM5, Arduino, HIL, system identification, sensor integration, flight testing